MIL-R-10509/14(USAF)
25 July 1969
SUPERSEDING
MIL-R-10509/006E(USAF)
12 July 1967

MILITARY SPECIFICATION

RESISTORS, FIXED, FILM (HIGH STABILITY)
STYLE RN80---TX

INACTIVE FOR NEW DESIGN

1. SCOPE

1.1 This specification covers the detail requirements for high-stability, film, fixed resistors designated Style RN80---TX. These resistors are required to comply with specific additional quality assurance provisions not required for conventional RN80 resistors. Characteristics B and G shall be applicable characteristics for the TX style resistors.

2. APPLICABLE DOCUMENTS

2.1 The following document, of the issue in effect on date of invitation for bids or request for proposal, forms a part of this specification to the extent specified herein.

SPECIFICATION

Military

MIL-R-10509 Resistors, Fixed, Film (High Stability), General Specifications for

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General specification. The requirements of MIL-R-10509 apply as requirements of this specification. When the two specifications conflict, this specification shall govern.

FSC 5905

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3.2 Lead material and finish. When weldable leads or other specific materials not in accordance with MIL-R-10509 are required, they shall be specified in the contract or purchase order (6.2).

3.3 Design and construction. The resistors shall be of the design, construction, and physical dimensions specified on figure 1.

3.4 Power rating. The power rating shall be 2 watts.

3.5 Voltage rating. The maximum continuous working voltage shall not exceed 750 volts.

3.6 Resistance. Resistance limits for the individual resistor characteristics shall be as follows:

Minimum Resistance	Maximum Resistance
Characteristic:	Characteristic:
B, C---30.1 ohms	B, G---10.0 megohms

3.7 Power conditioning. All resistors shall be power conditioned (100 percent inspection (see 4.4)). Resistance of all units shall be measured and shall be within specified tolerances before and after conditioning, and shall not have changed more than 0.5 percent.

3.8 Marking. Marking shall be in accordance with MIL-R-10509 with the addition of the letters TX placed immediately after the letter that indicates the resistance tolerance. For example:

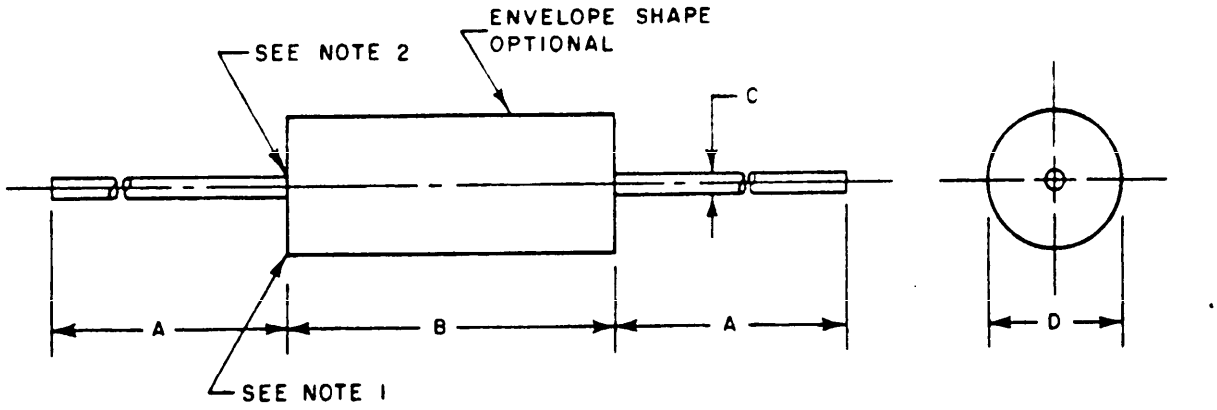
RN80
G
1003
CTX.

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection shall be in accordance with MIL-R-10509 and as specified herein.

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-R-10509 on the respective non-TX version. (The TX version shall be procured from sources listed in QPL-10509 for the respective non-TX version.)

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DIMENSIONS			
SYMBOL	INCHES	MILLIMETERS	
		MIN	MAX
A	1-1/2 ± 1/8	34.93	41.28
B	2-3/16 (+3/32, -1/16)	54.97	57.94
C	0.032 ± 0.002	0.762	0.863
D	3/8 (+1/16, -1/32)	8.73	11.11

NOTE:

1. The end of the body shall be that point at which the body diameter equals the nearest drill size larger than:

   a. 150 percent of the nominal lead diameter, or
   b. 300 percent of the nominal lead diameter.

   The same drill size shall be used for each end of the body. Either measuring system a or b shall be used except the system selected shall be used for all resistors measured.

2. The leads shall be solderable to within 0.062 inch (1.575 mm) of the resistor body.

FIGURE 1. Style RN80---TX Resistor

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4.3 Quality conformance inspection

4.3.1 Inspection of product for delivery. Inspection of product for delivery shall consist of groups A, B, and C inspections. Delivery of shipment shall not be delayed pending completion of group C testing.

4.3.2 Group A inspection. Group A inspection shall be in accordance with MIL-R-10509, except that power conditioning (4.4) shall be performed on each resistor prior to the tests listed and the AQL shall be 2.5 percent for minor defectives.

4.3.3 Group C inspection. Group C inspection shall be in accordance with MIL-R-10509. Successful performance of Group C inspection for the respective non-TX version will suffice to demonstrate conformance to this requirement for the TX version.

4.4 Power conditioning

4.4.1 Measurements before conditioning. Before power conditioning, the resistors shall have the d-c resistance measured at room conditions.

4.4.2 Conditioning procedures. Each resistor shall be subjected continuously to 150 percent rated wattage for a minimum of 12 hours under room temperature, free air conditions. Air circulation by normal convection is permitted. The applied rms voltage shall not exceed the maximum value specified (3.5).

4.4.3 Measurements after conditioning. After power conditioning, the resistors shall be stablized at room conditions for a minimum of 2 hours before resistance measurements are made. Resistors failing to meet the requirements of 3.7 shall be rejected. When the quantity rejected exceeds 5 percent of the lot, the entire lot shall be rejected. The supplier shall have a complete record available of all test data obtained during power conditioning testing (6.2). This data shall be maintained for a minimum of three years.

4.6 Terminal strength. Direct load shall be applied gradually until the load reaches 5 pounds.

5. PREPARATION FOR DELIVERY

5.1 Preparation for delivery shall be in accordance with MIL-R-10509.

6. NOTES

6.1 The notes specified in MIL-R-10509 and herein are applicable to this specification.

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6.2 Ordering data. Procurement documents should specify the following:

a. Requirement for weldable leads or other specific materials (3.2).

b. Whether power conditioning test data is to be supplied (4.4.3).

Custodian:
Air Force - 11

Review activities:
Air Force - 17, 85

Preparing activity:
Air Force - 11

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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